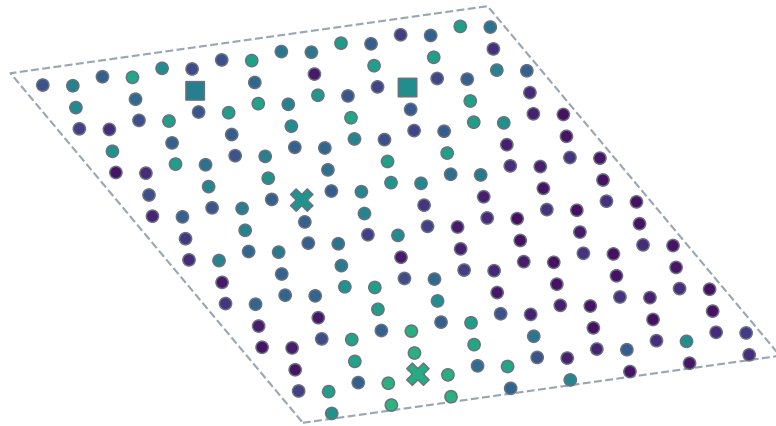


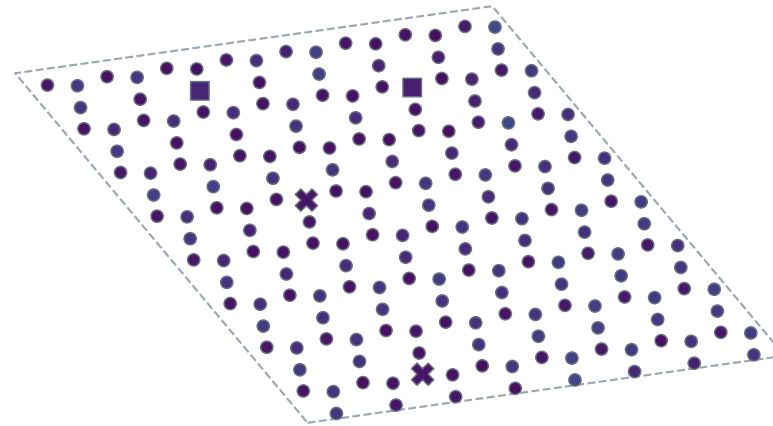
MoS2 | 4 defects | Mixed-low training
DFT target 2.5926 eV/defect; site IDs are zero-based

Attention
Prediction 2.8053 eV/defect | error 212.7 meV



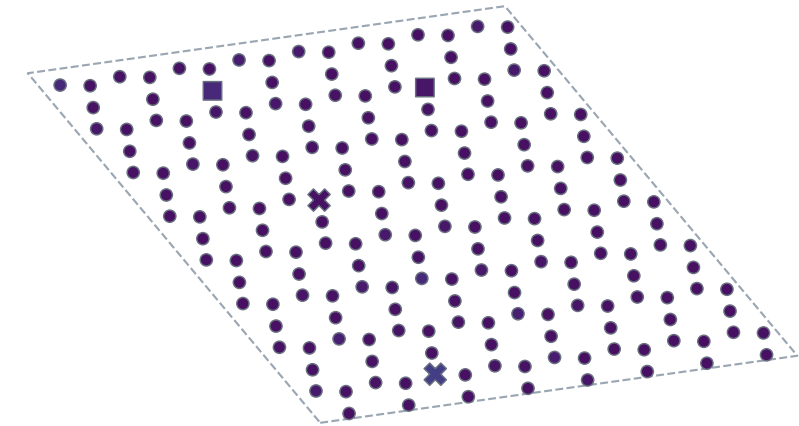
Soft-mask error 0.42 meV
Seed rank correlation 0.94; top-10% Jaccard 0.52

Independent hetero
Prediction 2.5936 eV/defect | error 1.0 meV

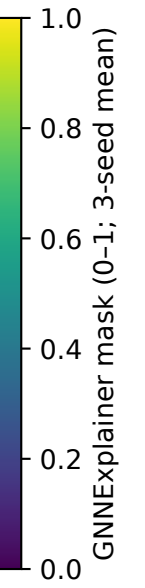


Soft-mask error 5.24 meV
Seed rank correlation 0.84; top-10% Jaccard 0.21

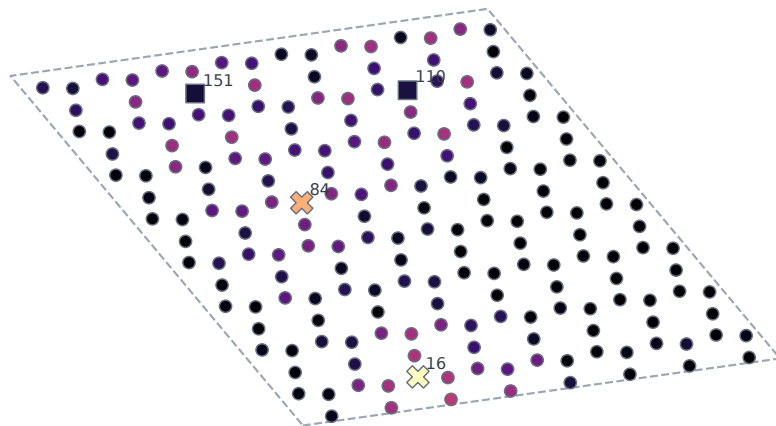
Shared hetero
Prediction 2.5888 eV/defect | error 3.8 meV



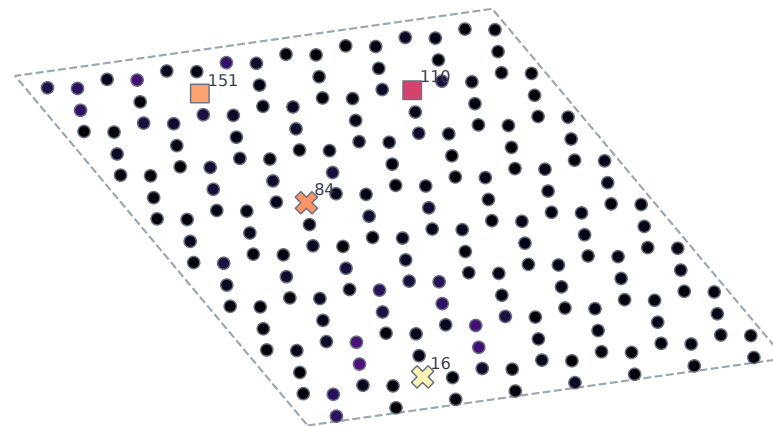
Soft-mask error 1.26 meV
Seed rank correlation 0.58; top-10% Jaccard 0.94



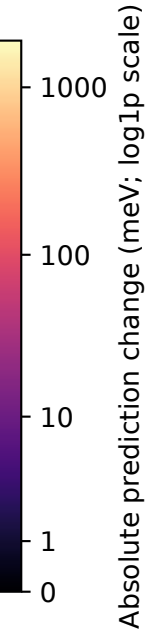
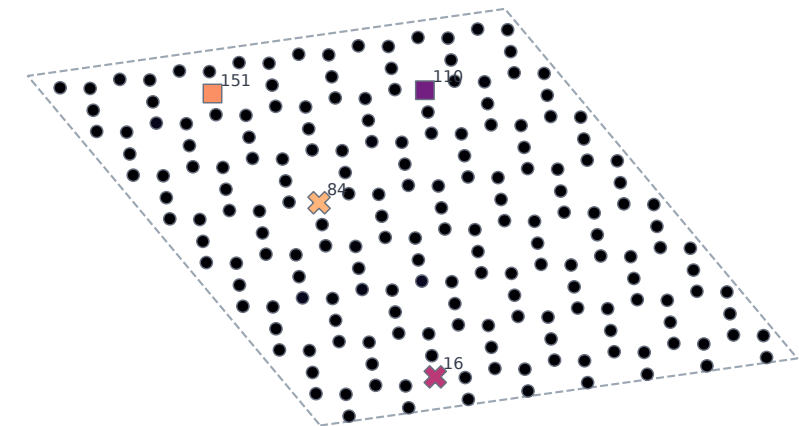
Single-site embedding ablation



Single-site embedding ablation



Single-site embedding ablation



● Pristine ■ Substitution ✕ Vacancy placeholder